

FCE 142: Lesson 1

Introduction to Engineering Drawings

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To confer the gift of drawing, we must create an eye that sees, a hand that obeys, a soul that feels; and in this task, the whole life must cooperate. In this sense, life itself is the only preparation for drawing. Once we have lived, the inner spark of vision does the rest.

Maria Montessori



1. Objectives

The main objectives of Engineering Drawings

◆ Objectives of the Introduction

- Describe the role of drawings in the design process
- List professionals in the local industry that use Technical Drawings
- Describe the design stages in civil engineering practice
- Explain why standards are important
- Identify use of graphic language



2. Overview

An engineering drawing, a type of technical drawing, is used to fully and clearly define requirements for engineered items.

Overview

- Engineering drawing (the activity) produces engineering drawings (the documents). It is a graphical language that communicates ideas and information from one mind to another. Most especially, it communicates all needed information from the engineer who designed a part to the workers who will make it.
- Graphic presentation is a basic, natural form of communication that is not tied to a particular time or place. It is in a sense, a universal language.
- Accomplishing ideas requires team work. A new product, machine, structure, or system may exist in the mind of an engineer or designer, but to become reality, it must be communicated to many different people. The ability to communicate design concepts quickly and accurately through technical drawings is key to meeting project budgets and time constraints. In the global market place, graphic communication is also an advantage where team members may not always share a common language.
- Graphic communication is the tool of trade for architects, engineers, designers, manufacturers, contractors, technicians, etc. Being an effective graphic communicator, by being able to represent ideas, designs and specifications in a consistent way, ensures that the product, system or structure that is envisioned is produced as specified.



3. Role of Technical Drawings

Drawings and specifications, control the many details of product manufacturing, assembly, structural and civil works, architectural works, maintenance, etc.

Understanding the Role of Engineering Drawings

- Sketches, 2D CAD drawings and 3D CAD models are all forms of technical drawings. Each of these types of drawings has a place in the process of designing and building a product, system or structure.
- Technical drawings can take the form of: Idea or concept sketches, Computation sketches, Design sketches, Layout drawings, Layout part drawings, Working or construction drawings, Electrical drawings, Installation Drawings, Assembly Drawings, et al.
- In general, technical drawings serve one of three purposes; Visualisation, Communication and Documentation
- Both ease in free hand sketching and use of CAD are valued skills in the global market place.
- Some examples of professions that use technical drawings to communicate and document designs include; civil and structural engineering, mechanical engineering, electrical engineering, architecture, landscape architecture, industrial design, etc.



4. Artistic & Technical Drawings

Graphic presentation has developed along 2 distinct lines of Artistic and Technical drawings

Artistic and Technical Drawings

- Humans have always used artistic drawings to express *aesthetic*, *philosophic* or other *abstract* ideas. Prior to the existence of other forms of communication, people informed themselves through conversation and by looking at sculptures, paintings, or drawings in public places. Pictures were a principle source of information that everyone could understand.
- Technical drawings are the other form of distinct form of drawings. Recorded history reveals that people have used drawings to represent the design of objects to be built or constructed
- Therefore whereas the purpose of artistic drawing is to convey emotion or artistic sensitivity in some way (*subjective impressions*), the purpose of engineering drawing is to convey information (*objective facts*).
- One of the corollaries that follows from this fact is that, whereas anyone can appreciate artistic drawing (*even if each viewer has his own unique appreciation*), engineering drawing requires some training to understand; but there is also a high degree of objective commonality in the interpretation.
- In fact, engineering drawing has evolved into a language that is more precise and unambiguous than natural languages; in this sense it is closer to a programming language in its communication ability. Engineering drawing uses an extensive set of conventions to convey information very precisely, with very little ambiguity.



5. The Design Process

This is the *organized* and *orderly* approach to solving problems. The engineering design process addresses society's needs, desires and problems by applying scientific principles, experience and creativity.

The Design Process

- Different types of technical drawings have a specific function in the engineering design process. The generic design process is as follows;

Problem identification

First, a clear statement of the need for and objectives for the design.

Ideation

Technical sketches are often used to convey the concepts to multi-disciplinary teams.

Refinement/Analys is

Designs may be rethought, based on engineering analysis. CAD models and sketches are useful during the analysis and compromise stage. Accurate 2D or 3D CAD models and drawings are created to refine design.

Implementation/ documentation

Production and/or working drawings providing the details of manufacture and assembly are finalised and approved.



6. Drafting Standards

These are standards that support a uniform, effective graphic language for use in the industry, manufacturing, engineering and science.

Drafting Standards

- These drafting standards are frequently updated to communicate information so that it meets the needs of modern industry and engineering practices. Standardization also aids internationalization, because people from different countries who speak different languages can read the same engineering drawing, and interpret it the same way. To that end, drawings should be as free of notes and abbreviations as possible so that the meaning is conveyed graphically.
- There are two main widely used standards;
 - a. International Standards – this is defined by the International Organisation for Standardisation (ISO)
 - b. American Standards – providing these standards has been the work of American National Standards Institute (ANSI) with the American Society for Engineering Education (ASEE), the Society of Automotive Engineers (SAE) and the American Society of Mechanical Engineers (ASME).
- The main differences between the two are;
 - a. The preferred method of projection i.e. first angle versus third angle
 - b. The units of measurement in dimensioning.



7. Legal Instrument

An engineering drawing is a legal document (that is, a legal instrument), because it communicates all the needed information about "what is wanted" to the people who will expend resources turning the idea into a reality.

Legal Instrument

- Engineering drawings, as legal instruments are therefore part of a contract; the purchase order and the drawing together, as well as any ancillary documents (engineering change orders [ECOs], called-out specs), constitute the contract. Thus, if the resulting product is wrong, the worker or manufacturer is protected from liability - as long as they have faithfully executed the instructions conveyed by the drawing. If those instructions were wrong, it is the fault of the engineer.
- Because manufacturing and construction are typically very expensive processes (involving large amounts of capital and payroll), the question of liability for errors has great legal implications as each party tries to blame the other and assign the wasted cost to the other's responsibility. This is the biggest reason why the conventions of engineering drawing have evolved over the decades toward a very precise, unambiguous state.



8. Media

For centuries, until the post-World War II era, all engineering drawing was done manually by using pencil and pen on paper or other substrate (e.g., vellum, mylar). However, since the advent of computer-aided design (CAD), engineering drawing has been done more and more in electronic medium with each passing decade. Today most engineering drawing is done with CAD, although pencil and paper have not disappeared.

Media

- Some of the tools of manual drafting include pencils, pens and their ink, straightedges, T-squares, French curves, triangles, rulers, protractors, dividers, compasses, scales, erasers, and tacks or push pins. (Slide rules used to number among the supplies, too, but nowadays even manual drafting, when it occurs, benefits from a pocket calculator or its onscreen equivalent.) And of course the tools also include drawing boards (drafting boards) or tables. The English idiom "to go back to the drawing board", which is a figurative phrase meaning to rethink something altogether, was inspired by the literal act of discovering design errors during production and returning to a drawing board to revise the engineering drawing. Drafting machines are devices that aid manual drafting by combining drawing boards, straightedges, pantographs, and other tools into one integrated drawing environment. CAD provides their virtual equivalents.
- Producing drawings usually involves creating an original that is then reproduced, generating multiple copies to be distributed to the shop floor, vendors, company archives, and so on. The classic reproduction methods involved blue and white appearances (whether white-on-blue or blue-on-white), which is why engineering drawings were long called, and even today are still often called, "blueprints" or "bluelines", even though those terms are anachronistic from a literal perspective, since most copies of engineering drawings today are made by more modern methods (often inkjet or laser printing) that yield black or multicolour lines on white paper. The more generic term "print" is now in common usage to mean any paper copy of an engineering drawing. In the case of CAD drawings, the original is the CAD file, and the printouts of that file are the "prints".



9. Design Process in Portfolio

A portfolio is a representative sample of work that helps communicate your skills and talent, usually to a prospective employer or client (and in this case, examiner)

Design Process in Portfolio

Showing an example of your work offers an advantage over just talking about it. Drawings are proof that you not only have good ideas but also have the skill to communicate clearly and accurately throughout the stages of project, from concept to completion.

Some tips for creating a successful portfolio;

- Less is more. A few good examples will do. Quality over quantity.
- Show correct drawing practices. Update a portfolio as you increase knowledge on technical drawing standards.
- Keep several versions of your portfolio, including hard copy, digital and even web presence.
- Never rely entirely on electronic equipment.
- Save good examples as you create them.
- Depending on career goals, let your portfolio reflect the area you want to explore.



10. Assignments

Assignments will form the content of the portfolio

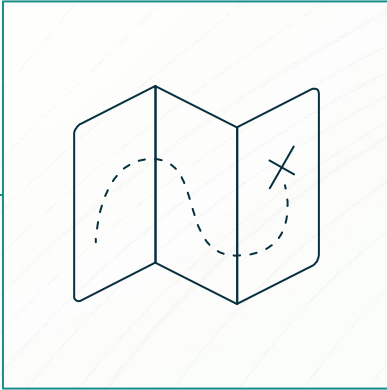
Assignments

ASSIGNMENT 1

- Attach examples of each of the following types of Technical drawings to your portfolio: Idea or concept sketches, Computation sketches, Design sketches, Layout drawings, Layout part drawings, Working or construction drawings, Electrical drawings, Installation Drawings and Assembly Drawings.

ASSIGNMENT 2

- List the design stages, with a short description, of a civil engineering project. Attach to your portfolio.



Thanks!

Any questions?

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